

Tuning Electrogenerated Chemiluminescence Intensity Enhancement Using Hexagonal Lattice Arrays of Gold Nanodisks

Thomas S. Heiderscheit,[¶] Shunpei Oikawa,[¶] Stephen Sanders, Hiro Minamimoto, Emily K. Searles, Christy F. Landes,^{*} Kei Murakoshi,^{*} Alejandro Manjavacas,^{*} and Stephan Link^{*}



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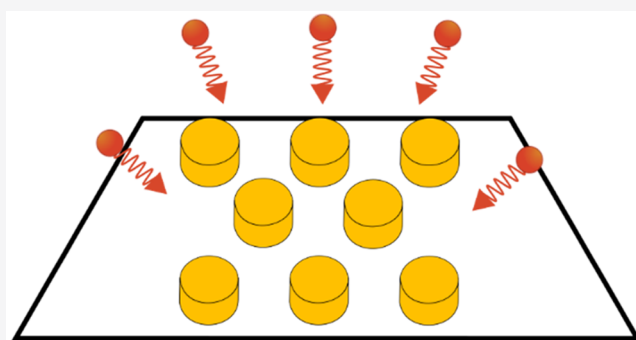


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Supporting Information

ABSTRACT: Electrogenerated chemiluminescence (ECL) microscopy shows promise as a technique for mapping chemical reactions on single nanoparticles. The technique's spatial resolution is limited by the quantum yield of the emission and the diffusive nature of the ECL process. To improve signal intensity, ECL dyes have been coupled with plasmonic nanoparticles, which act as nanoantennas. Here, we characterize the optical properties of hexagonal arrays of gold nanodisks and how they impact the enhancement of ECL from the coreaction of tris(2,2'-bipyridyl)dichlororuthenium(II) hexahydrate and tripropylamine. We find that varying the lattice spacing results in a 23-fold enhancement of ECL intensity because of increased dye-array near-field coupling as modeled using finite element method simulations.



Electrogenerated chemiluminescence (ECL) microscopy is a promising technique for monitoring chemical activity on single nanoparticles. ECL is the emission of light using electrochemically driven redox reactions.^{1,2} As a luminescence source that does not require external optical excitation, ECL provides a low background relative to fluorescence and Raman microscopy. Additionally, dyes are regenerated after emission, eliminating the problems caused by photobleaching seen in fluorescence microscopy.^{1,2} One of the current limitations of ECL arises from the technique's low spatial resolution. ECL microscopy has only demonstrated submicron spatial resolution³ because of the low quantum yield of ECL^{1,4} and the diffusive nature of the ECL dye.^{3,5} This spatial resolution is far below techniques like fluorescence microscopy, motivating research into methods for enhancing ECL intensity.

One avenue for enhancing ECL intensity is coupling ECL dyes with gold nanoparticles.^{3,5–12} The interaction of the dye emission with the nanoparticle's coherent oscillation of conduction band electrons, or localized surface plasmon resonance (LSPR), has been demonstrated to both enhance and quench ECL.^{7,9} Previous studies have revealed that at short distances (1–5 nm),^{7,9,13} emission is quenched because of energy transfer from the dye to the nanoparticle. At larger distances (5–20 nm),^{7,9} emission is enhanced because of a LSPR-induced modulation of the local density of optical states.^{10,14–18} Enhancement can be further improved using large electric fields at sharp tips¹⁹ or in the junction of two or more nanoparticles.^{20,21} Additionally, the impact of nanoparticle size,²² shape,⁸ material composition,^{6,12} local environ-

ment,⁵ and spectral overlap¹⁰ on ECL intensity has been investigated.

Although patterned arrays of nanoparticles have been used for ECL studies,^{23–25} no study, thus far, has investigated the role plasmonic nanoparticle arrays play in ECL enhancement. Periodic nanoparticle arrays display unique spectral features arising from the coherent superposition of the individual response of their constituents. Furthermore, when the coupling between elements of the array is sufficiently large to establish long-range interactions between them, these systems can support collective modes known as surface lattice resonances (SLRs).^{26–31} These resonances are highly tunable^{32–34} with line widths that are narrower than those observed for single nanoparticles.^{30,31} SLRs have demonstrated fluorescence enhancement through the increase of the radiative component of the local density of optical states³⁵ and the modification of the emitter's radiation pattern to improve the collection efficiency.³⁶ While the excitation rate for ECL is dependent on electrochemical potential, the emission of ECL dyes could be impacted similarly to fluorescence, leading to strong enhancement by periodic arrays of nanostructures.

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This work demonstrates the tuning of the ECL intensity enhancement with periodic arrays of metallic nanostructures by measuring the emission of the tris(2,2'-bipyridyl)-dichlororuthenium(II) ($\text{Ru}(\text{bpy})_3^{2+}$) and tripropylamine (TPA) coreactant pair at the surface of hexagonally packed arrays of gold nanodisks. The optical properties of the arrays were first characterized, both experimentally and theoretically, to understand the dependence of the array's spectral features on nanodisk spacing and measurement geometry. The spectrum and signal intensity of $\text{Ru}(\text{bpy})_3^{2+}$ were then measured for all lattice spacings, and finite element method (FEM) simulations were used to replicate the observed trends seen in the ECL intensity enhancement.

Hexagonally packed arrays had differences in observed color because of their varying nanoparticle spacing. Gold nanodisks, 100 nm in diameter and height, were lithographically fabricated into a $35\ \mu\text{m} \times 35\ \mu\text{m}$ hexagonal lattice array, as illustrated in Figure 1a. The spacing between individual nanodisks, a , was varied from 200 to 575 nm. Figure 1b displays representative SEM micrographs of the arrays with $a = 200$ nm (green), 300 nm (magenta), 400 nm (blue), and 500 nm (red). The sample was designed and fabricated to allow

simultaneous measurement of each array covering all lattice spacings. Figure 1c gives a digital camera image of all arrays observed in a microscope. Changing a resulted in a change of the array's observed color, with orange, purple, teal, and salmon observed for $a = 250, 350, 450$, and 550 nm, respectively.

The spectral features of the arrays were tuned by varying the measurement geometry, local environment, and lattice spacing. Figure 2 provides a general characterization of the influence of

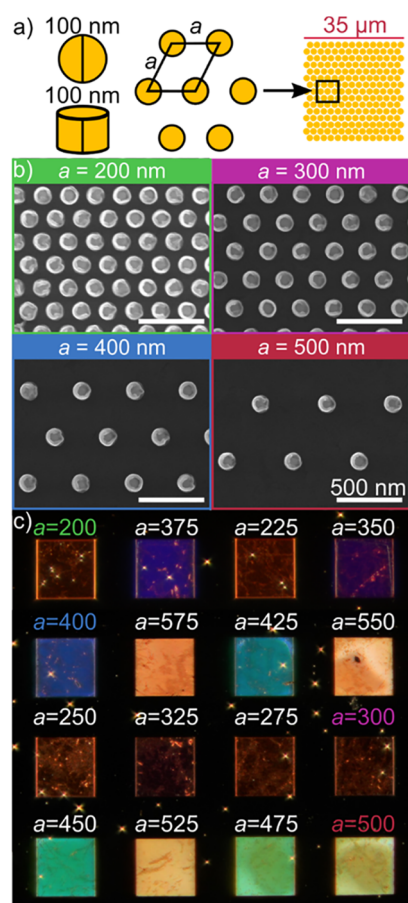


Figure 1. Physical characterization of gold nanodisk arrays. (a) Schematic of array geometry. Sixteen $35\ \mu\text{m} \times 35\ \mu\text{m}$ arrays were fabricated per sample, consisting of 100 nm diameter and height nanodisks hexagonally arranged with a lattice spacing, a , ranging from 200 to 575 nm in steps of 25 nm. (b) SEM images of arrays with 200 nm (green), 300 nm (magenta), 400 nm (blue), and 500 nm (red) lattice spacings. Scale bars correspond to 500 nm. (c) Digital camera image of all arrays.

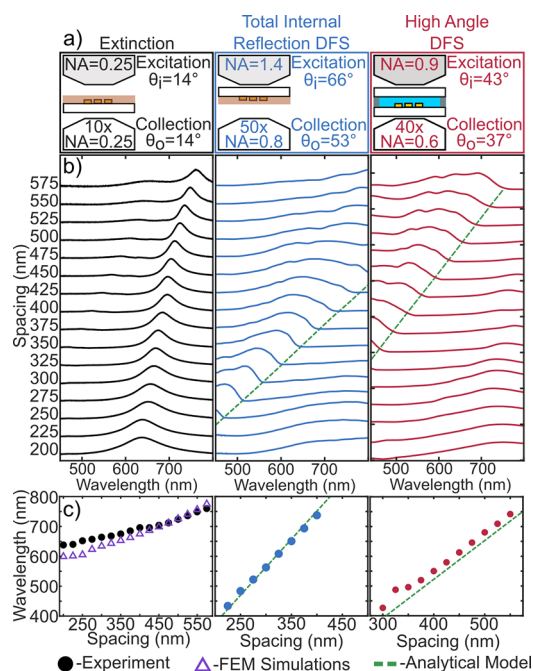


Figure 2. Theoretical and experimental optical characterization of gold nanodisk arrays embedded in polymer. (a) Schematics depicting the extinction (black), TIRDFS (blue), and HADFS (red) measurement geometries used. (b) Spectra for all arrays of varying lattice spacings acquired with the extinction (black), TIRDFS (blue), and HADFS (red) geometries. The dashed green lines for the TIRDFS and HADFS geometries indicate the theoretical position of the long-wavelength cutoff for constructive interference. TIRDFS and HADFS spectra were calculated using the method described in Figure S1. (c) Comparison of the predicted and experimental resonance wavelengths and long-wavelength cutoff for constructive interference (center and right only) for the extinction (black), TIRDFS (blue), and HADFS (red) geometries, respectively. FEM simulation predicted resonance wavelengths are represented by purple triangles, and analytical modeling of the long-wavelength cutoff are given by green dashed lines. The model displayed good agreement with r^2 values of 0.98 and 0.72 for the TIRDFS and HADFS spectral measurements, respectively.

excitation and collection conditions. Figure 2a displays the excitation and collection geometries used for extinction (black), total internal reflection dark-field scattering (TIRDFS) (blue), and high angle dark-field scattering (HADFS) (red) measurements. Excitation of the array and collection of scattered light at non-normal angles resulted in changes to the measured spectra because the arrays behaved as diffraction gratings.^{33,34}

Figure 2b summarizes the spectra of all lattice spacings for extinction (left), TIRDFS (center), and HADFS (right) geometries for arrays coated in a polymer to ensure the same conditions as the ECL measurements discussed below.

As a increased from 200 to 575 nm, the extinction resonance wavelength red-shifted and the line width narrowed. Similar spacing-dependent trends have been experimentally and theoretically presented in the literature.^{32,37} Varying a therefore weakly tuned the resonance wavelength from 640 to 760 nm in extinction. For near-normal incidence excitations, such as the excitation cone spanning 14° used for these measurements, the refractive indices of the substrate and superstrate must match to observe SLRs.³⁸ The refractive index mismatch of PVA (~ 1.47) and indium tin oxide (ITO, 1.7) partially hindered the long-range interaction between the individual nanoparticles required for sustaining strong SLRs. Therefore, the nanodisks interacted weakly with their neighbors, resulting in the limited resonance wavelength tunability observed in the extinction measurements.³⁸

A larger resonance wavelength tunability was achieved by changing the excitation and collection conditions to the dark-field configurations of TIRDFS and HADFS. As opposed to the extinction measurements, under these configurations the array acted as a diffraction grating, resulting in an optical response strongly dependent on the array periodicity. Specifically, the TIRDFS and HADFS measurement geometries, which used 66° and 43° excitation angles, respectively, gave a similar increase in resonance wavelength as a increased, covering a spectral range of ~ 300 nm. The resonance wavelength tunability for the TIRDFS geometry was found to be minimally impacted by changing the refractive index of the superstrate (Figure S2), when switching from a polymer to an air environment.

The spectroscopic characteristics of the arrays were well reproduced using simulations and analytical modeling, reinforcing the experimentally observed influence of excitation and collection conditions on the measured spectra. FEM simulations of the extinction measurement geometry were performed (Figure S3) and compared to measured extinction spectra. The left panel of Figure 2c compares the resonance wavelength of the experimental (black circle) and FEM simulated (purple triangle) spectra for each lattice spacing, showing good agreement. The spectra for the TIRDFS and HADFS measurement geometries in Figure 2b can be modeled analytically considering conservation of momentum across the system. Because of the periodicity of the array, the parallel components of the incident, $\vec{K}_{\text{in}}^{\parallel}$, and emitted, $\vec{K}_{\text{out}}^{\parallel}$, wavevectors are related by

$$\vec{K}_{\text{out}}^{\parallel} = \vec{K}_{\text{in}}^{\parallel} + \vec{q}_{ij} \quad (1)$$

where $\vec{q}_{ij}$ is ij -th reciprocal lattice vector of the array. For the hexagonal array depicted in Figure 1a (taking the x -axis as the horizontal one), $\vec{K}_{\text{in}}^{\parallel}$, $\vec{K}_{\text{out}}^{\parallel}$, and $\vec{q}_{ij}$ are defined below as follows:

$$\vec{K}_{\text{in}}^{\parallel} = \frac{2\pi}{\lambda} \text{NA}_{\text{in}} [\cos(\phi_{\text{in}})\hat{x} + \sin(\phi_{\text{in}})\hat{y}] \quad (2)$$

$$\vec{K}_{\text{out}}^{\parallel} = \frac{2\pi}{\lambda} \text{NA}_{\text{out}} [\cos(\phi_{\text{out}})\hat{x} + \sin(\phi_{\text{out}})\hat{y}] \quad (3)$$

$$\vec{q}_{ij} = \frac{2\pi}{a} \left[i\hat{x} + \frac{(2j-i)}{\sqrt{3}}\hat{y} \right] \quad (4)$$

where λ is the wavelength of light, NA_{in} and NA_{out} are the incident and collection numerical apertures, and ϕ_{in} and ϕ_{out} are the in-plane azimuthal angles for the incident and collected

wavevectors, respectively. Then, for each $\vec{q}_{ij}$, eq 1 determines the range of wavelengths for which constructive interference between the radiation emitted by the array is possible. As can be readily seen from eq 4, the six smallest reciprocal lattice vectors ($\vec{q}_{\pm 1,0}$, $\vec{q}_{0,\pm 1}$, and $\vec{q}_{\pm 1,\pm 1}$) all have the same length $4\pi/\sqrt{3}a$, and therefore, following eq 1, they determine the cutoff wavelength, λ_{max} , above which constructive emission is not possible. In particular, exploiting the axial symmetry of the experiment, λ_{max} can be written as

$$\lambda_{\text{max}} = a \frac{\sqrt{3}}{2} [\text{NA}_{\text{in}} + \text{NA}_{\text{out}}] \quad (5)$$

This expression was used to predict the long-wavelength cutoff observed for the TIRDFS and HADFS measurement geometries. In Figure 2b, the analytical model is overlaid with the TIRDFS and HADFS spectra. The long-wavelength cutoffs were determined for the TIRDFS and HADFS measurement geometries to compare the analytical model more quantitatively to the experimental data (Figure S4). Those experimental results are plotted in Figure 2c together with the analytical model (green dashed line), demonstrating excellent agreement. The array's spectra could therefore be easily predicted for different values of a , NA_{in} , and NA_{out} . Note that this model also predicts the observed refractive index independence (Figure S2).

Experimental measurements indicated $\text{Ru}(\text{bpy})_3^{2+}$ couples to the arrays, resulting in spacing-dependent ECL intensity enhancement. The enhancement for each lattice spacing was calculated (Figure S5) as the ratio of the ECL intensity from the array (I_{Array}) to the intensity from the bare ITO (I_{ITO}) for two separate samples (Figure S6). I_{Array} of each sample was measured simultaneously for all arrays with varying lattice spacings (Figure S5), ensuring that ECL currents were comparable. The ECL intensity enhancement was normalized for the total surface area of I_{Array} and I_{ITO} to account for the differences in the number of disks for each array (Figure S7). Increasing a with a constant array size resulted in fewer nanodisks, decreasing the available gold surface area for $\text{Ru}(\text{bpy})_3^{2+}$ and TPA to react and hence decreasing the measured ECL intensity.^{1,10} Figure 3a shows the lattice spacing dependence of ECL intensity enhancement after correcting for the total surface area. Initially, an increased lattice spacing led to an increase in ECL enhancement, as illustrated by the inset ECL images with $a = 225$ nm (green) and $a = 350$ nm (purple). A maximum enhancement of 23 times was achieved at $a = 375$ nm. A further increase in a caused a decrease in ECL enhancement, visualized by the inset ECL images for $a = 350$ nm (purple) and $a = 475$ nm (cyan).

$\text{Ru}(\text{bpy})_3^{2+}$ coupled weakly though to the arrays as indicated by the lattice spacing not influencing the shape of the $\text{Ru}(\text{bpy})_3^{2+}$ spectrum. Figure 3b shows the $\text{Ru}(\text{bpy})_3^{2+}$ emission spectra measured on the arrays for all lattice spacings (Figure S8). Like the changes in ECL enhancement, the emission spectra initially increased in intensity with increased lattice spacing. ECL intensity then decreased for spacings between 475 and 575 nm. While the spectral intensity varied with lattice spacing, consistent with the analysis in Figure 3a based on ECL images, the overall spectral shape did not change. This result revealed that $\text{Ru}(\text{bpy})_3^{2+}$ only weakly coupled with the arrays, as intermediate or strong coupling would have resulted in peak splitting.^{40–43} Strong coupling is

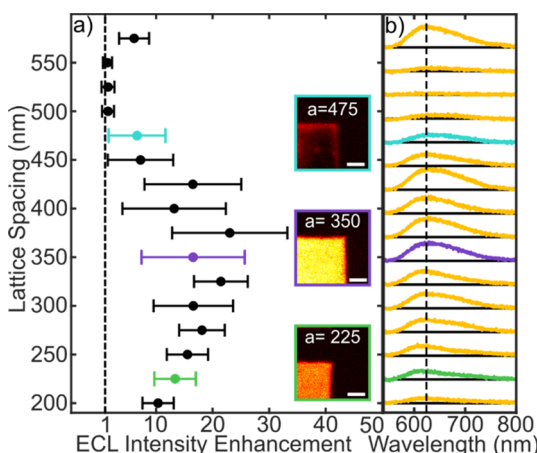


Figure 3. Comparison of spectra and ECL enhancement of arrays with varying lattice spacings. (a) Lattice spacing dependent ECL intensity enhancement after correcting for the number of disks in the array. Corresponding ECL images for arrays with 225 nm (green), 350 nm (purple), and 475 nm (cyan) lattice spacings are displayed as insets. Scale bars correspond to 10 μm . Intensity enhancement was calculated using the method presented in Figures S5 and S6. (b) $\text{Ru}(\text{bpy})_3^{2+}$ spectrum collected from all arrays. The black dashed line indicates 624 nm, the emission maximum of $\text{Ru}(\text{bpy})_3^{2+}$.

limited by the incoherent excitation exhibited by the $\text{Ru}(\text{bpy})_3^{2+}$ dyes.

The general optical characterization of the arrays, presented in Figure 2, was not sufficient to replicate how $\text{Ru}(\text{bpy})_3^{2+}$ excited the arrays or to explain the trends in ECL intensity enhancement. While ECL intensity has been previously reported to improve when the nanoparticle-dye spectral overlap was maximized,¹⁰ a simple analysis comparing the ECL spectrum with the array resonances does not reproduce the observed trend in ECL enhancement. Such a comparison is not valid here because the measurement geometries used in Figure 2 did not replicate how $\text{Ru}(\text{bpy})_3^{2+}$ excited the arrays. Specifically, for ECL measurements, the arrays were excited in the near-field by the $\text{Ru}(\text{bpy})_3^{2+}$ dyes. The dyes acted as independent point dipole emitters, resulting in an incoherent excitation that limited the array's collective behavior.^{31,39} All of the measurement geometries discussed in Figure 2 excite the array coherently though and therefore could not fully explain the experimental trends, leading to the necessity of modeling the ECL enhancement.

To model the ECL measurements, the intensity of the radiation passing through the substrate was calculated for the array being excited by a localized dipole source simulating a $\text{Ru}(\text{bpy})_3^{2+}$ dye using FEM simulations (see Supporting Information for details). This calculation had to be repeated for enough positions of the dipolar source to obtain an averaged result over the region of interest since, in the experiment, the dyes were distributed all around the nanodisks. This region, as indicated in Figure 4a, could be modeled as a conformal shell around the nanodisk with a thickness s . However, instead of following this procedure, Lorentz reciprocity was exploited, which states that, for systems made of linear time-invariant materials with symmetric dielectric tensors, Maxwell's equations are invariant under the exchange of the source and observation points.^{44,45} In particular, the original problem involved a dipole source located near the nanodisk and an observation point in the far-field of the substrate side (left panel of Figure 4a). Therefore, the

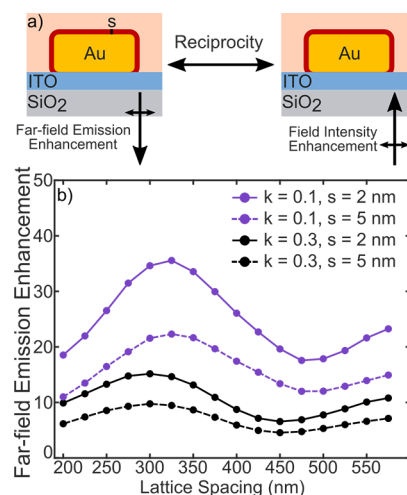


Figure 4. FEM simulations of the ECL intensity enhancement. (a) Schematics of the FEM simulation geometry for the system under study and its reciprocal version. (b) Lattice spacing dependent far-field emission enhancement calculated at a wavelength of 624 nm as a function of the shell thickness, s , and the imaginary part of the refractive index of the polymer layer, k .

reciprocal version of this problem consisted of a plane wave excitation coming from the substrate and an observation point, at which the electric field was calculated, corresponding to the location of the dipole source in the original system (right panel Figure 4a).⁴⁶ This procedure was advantageous because, due to the nature of the FEM calculations, it was possible to obtain the electric field at all mesh nodes in the simulation domain with a single simulation run, thus enabling an efficient calculation of the average over all possible positions of the $\text{Ru}(\text{bpy})_3^{2+}$ dyes within the region of interest. Notice that this advantage came at the expense of obtaining the intensity radiated only in one direction (i.e., the opposite direction of the plane wave excitation in the reciprocal problem). However, since the dominant mode of the nanodisks corresponded to a dipole oscillating parallel to the plane of the array and the collection angle of the experimental setup was 37° , this approximation was expected to be sufficiently accurate. Furthermore, the environment around the nanodisk was set to be a polymer layer containing the dyes, which we modeled using a refractive index $n + ik$, with $n = 1.43$ and varying imaginary part k . This imaginary part, which simulated an artificial loss, was introduced to effectively account for the fact that, in the ECL experiments, the $\text{Ru}(\text{bpy})_3^{2+}$ dyes behaved as incoherent emitters. Results were normalized relative to those simulated in the absence of the array.

These FEM simulations provided insight into the mechanism by which the $\text{Ru}(\text{bpy})_3^{2+}$ coupled to the arrays. Figure 4b illustrates the dependence of the far-field emission enhancement on a , calculated at 624 nm (the emission maximum of $\text{Ru}(\text{bpy})_3^{2+}$), for two values of k and s . In all cases, the trend of the enhancement matched the experimentally measured ECL intensity enhancement (Figure 3a), with an approximate maximum enhancement at $a \sim 300\text{--}350$ nm. The appearance of this maximum can be attributed to two main factors: (i) an optimal spectral overlap between ECL and the array's optical response, which led to $\text{Ru}(\text{bpy})_3^{2+}$ efficiently exciting the array, and (ii) an optical response that favored emission over absorption, thus helping to achieve a more efficient radiation to the far-field. Increasing the shell thickness from 2 nm (solid

line) to 5 nm (dashed line) resulted in a decrease in the far-field emission enhancement. An increased shell thickness placed dyes further away from the nanoparticle, resulting in a lower plasmonic enhancement of ECL.^{7,9} Increasing k from 0.1 (purple) to 0.3 (black), which effectively reduced the collective response of the dyes coupled to the array, resulted in a decrease of the far-field emission enhancement for both shell thicknesses. These results, which could not be obtained using the extinction, TIRDFS, and HADFS measurements, indicated that both the location of the dyes and the coherence of ECL play a large role in ECL enhancement.

In summary, we studied the enhancement of ECL by coupled periodic arrays of plasmonic nanoparticles. Three measurement geometries, along with modeling, were used to characterize the optical properties of hexagonal lattice arrays of gold nanodisks. Resonance wavelength tunability was shown to be impacted by measurement geometry, local environment, and lattice spacing. The ECL intensity and spectrum were measured for all lattice spacings. A 23-fold ECL intensity enhancement and no change in the $\text{Ru}(\text{bpy})_3^{2+}$ spectrum indicated that $\text{Ru}(\text{bpy})_3^{2+}$ dyes were weakly coupled to the arrays. FEM simulations were needed to fully explain the near-field coupling between $\text{Ru}(\text{bpy})_3^{2+}$ emission and the plasmonic arrays because the optical characterization of the arrays did not sufficiently replicate how the dyes excited the arrays as incoherent point dipole sources in the near-field. These FEM simulations revealed that the ECL intensity enhancement was impacted by both the location of the dyes and the coherence of ECL. While the incoherence of dye emission cannot be controlled, enhancement was theoretically predicted to be improved by placing the dyes closer to the array. Restricting $\text{Ru}(\text{bpy})_3^{2+}$ to distances where the array can still enhance ECL, using, for example, a mesoporous shell^{47–50} or layer by layer deposition,⁵¹ could be utilized to improve the ECL intensity. This work therefore provides insight into the properties of ECL enhancement substrates that need to be considered when scaling from single nanostructures to ordered nanoparticle arrays.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcllett.0c03564>.

Experimental and numerical methods, calculation of the average TIRDFS and HADFS spectra, TIRDFS spectra in air and polymer medium, FEM extinction spectra, analysis method for determining the long wavelength cutoff for constructive interference, ECL intensity enhancement calculation, lattice/ITO surface available surface area calculation, and the method for obtaining ECL spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Christy F. Landes – Department of Chemistry, Department of Electrical and Computer Engineering, and Department of Chemical and Biomolecular Engineering, Rice University, Houston, Texas 77005, United States; orcid.org/0000-0003-4163-6497; Email: clandes@rice

Kei Murakoshi – Department of Chemistry, Faculty of Science, Hokkaido University, Sapporo, Hokkaido 060-0810, Japan;

orcid.org/0000-0003-4786-0115; Email: kei@sci.hokudai.ac.jp

Alejandro Manjavacas – Department of Physics and Astronomy, University of New Mexico, Albuquerque, New Mexico 87106, United States; Instituto de Óptica (IO-CSIC), Consejo Superior de Investigaciones Científicas, 28006 Madrid, Spain; orcid.org/0000-0002-2379-1242; Email: a.manjavacas@csic.es

Stephan Link – Department of Chemistry and Department of Electrical and Computer Engineering, Rice University, Houston, Texas 77005, United States; orcid.org/0000-0002-4781-930X; Email: slink@rice.edu

Authors

Thomas S. Heiderscheit – Department of Chemistry, Rice University, Houston, Texas 77005, United States

Shunpei Oikawa – Department of Chemistry, Faculty of Science, Hokkaido University, Sapporo, Hokkaido 060-0810, Japan

Stephen Sanders – Department of Physics and Astronomy, University of New Mexico, Albuquerque, New Mexico 87106, United States

Hiro Minamimoto – Department of Chemistry, Faculty of Science, Hokkaido University, Sapporo, Hokkaido 060-0810, Japan

Emily K. Searles – Department of Chemistry, Rice University, Houston, Texas 77005, United States

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.jpcllett.0c03564>

Author Contributions

[†](T.S.H., S.O.) Both authors contributed equally to this work.

Notes

The authors declare no competing financial interest.

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